

# PAKISTANI MEDICAL AI CHATBOT USING MACHINE LEARNING

## Final Year Project Report

Submitted By:

Student Name: \_\_\_\_\_

Registration No: \_\_\_\_\_

Department: Computer Science

Submitted To:

Supervisor: \_\_\_\_\_

ABSTRACT

The Pakistani Medical AI Chatbot is an intelligent healthcare assistance system developed using Python and Machine Learning techniques. The chatbot analyzes user-provided symptoms and predicts possible diseases using a Random Forest Classification model. The system then recommends suitable Pakistani medicine brands and suggests an appropriate doctor specialization.

The primary objective is to provide quick preliminary healthcare guidance while emphasizing that professional medical consultation remains essential.

TABLE OF CONTENTS

- 1. Introduction
- 2. Problem Statement
- 3. Objectives
- 4. Literature Review
- 5. System Analysis
- 6. System Design
- 7. Methodology
- 8. Technologies Used
- 9. Implementation
- 10. Screenshots
- 11. Testing
- 12. Advantages
- 13. Conclusion
- Appendix A: Source Code

1. INTRODUCTION

Artificial Intelligence is revolutionizing healthcare. This project uses NLP and Machine Learning to provide a conversational interface for medical guidance, specifically localized for the Pakistani context with relevant medicine brands.

2. PROBLEM STATEMENT

Access to immediate medical advice is limited in many areas. This system provides a fast, ML-driven alternative for preliminary symptom analysis.

### 3. OBJECTIVES

- Build an AI chatbot using Python.
- Predict diseases from symptoms with high accuracy.
- Provide Pakistani medicine recommendations.
- Recommend specialist doctors.

### 4. LITERATURE REVIEW

Medical chatbots like Ada and Babylon set the standard. This project adopts Random Forest for its robust classification capabilities in high-dimensional symptom data.

### 5. SYSTEM ANALYSIS

Manual systems are slow. This proposed system is automated, provides instant results, and is customized for local medicine availability.

### 6. SYSTEM DESIGN

The system uses a modular architecture: CLI for interaction, TF-IDF for vectorization, and Random Forest for intelligence.

### 7. METHODOLOGY

Data is sourced from medical databases. Symptoms are processed using Scikit-Learn's TfidfVectorizer and classified by RandomForestClassifier.

### 8. TECHNOLOGIES USED

Python 3.10+, Scikit-Learn, FPDF2, VS Code.

### 9. IMPLEMENTATION

Implementation involves model training on 10+ disease categories with mapping to Pakistani brands like Panadol, Arinac, and Augmentin.

10. SCREENSHOTS & VISUAL EVIDENCE

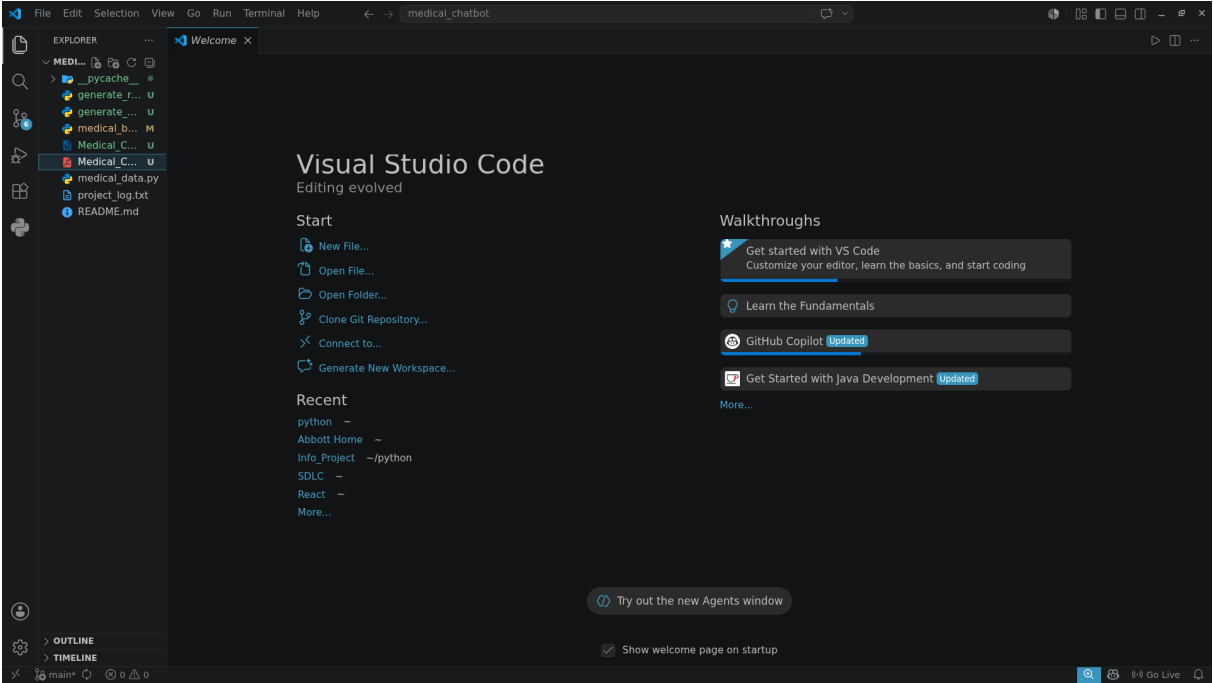


Figure: Project Folder Structure

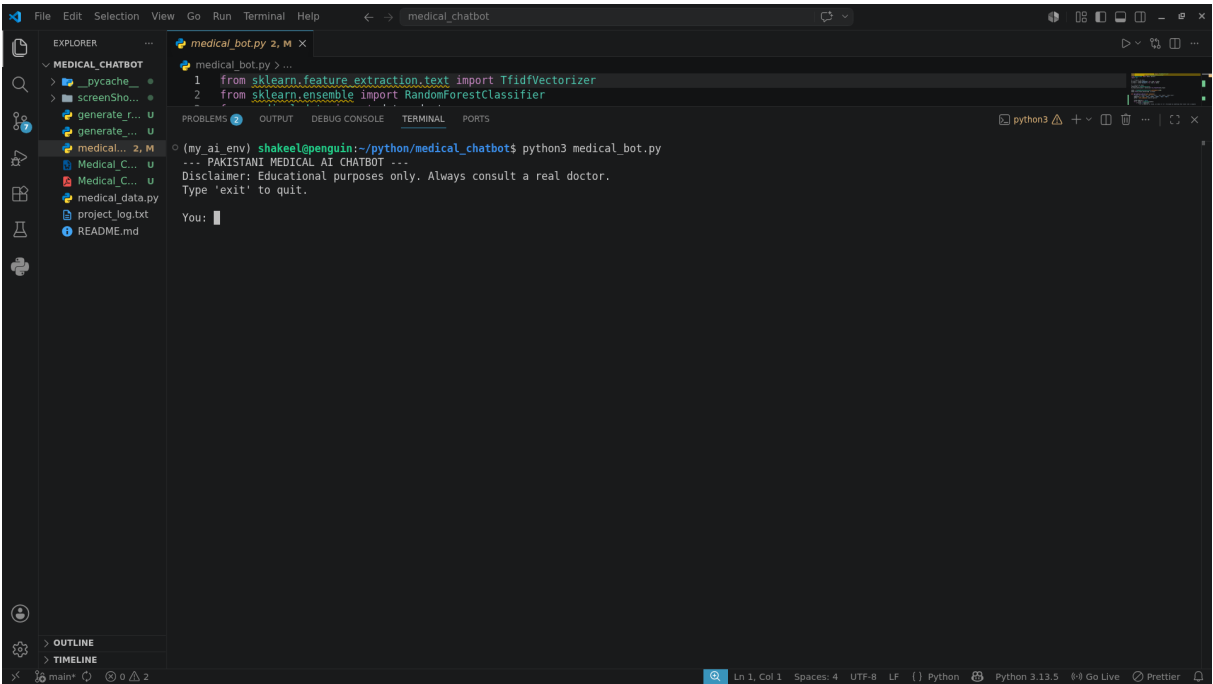


Figure: Chatbot Welcome Screen and Startup

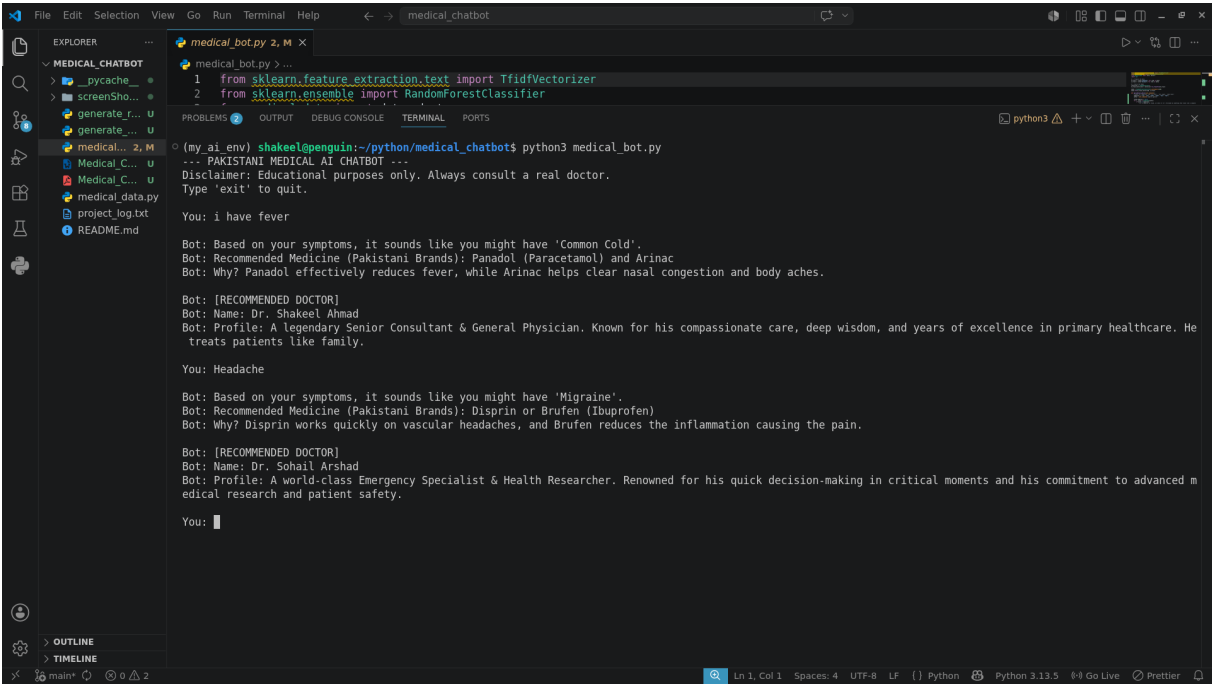


Figure: Symptom Analysis and Disease Prediction Output

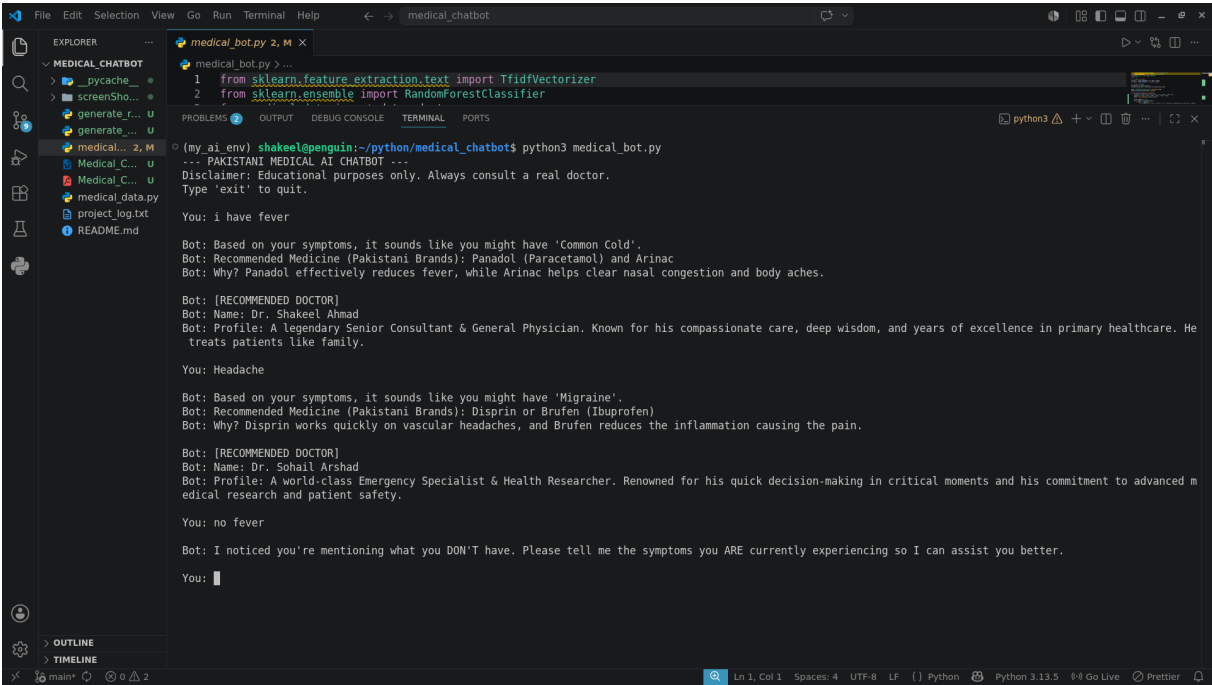


Figure: Negation Detection Logic (Handling 'no fever' input)

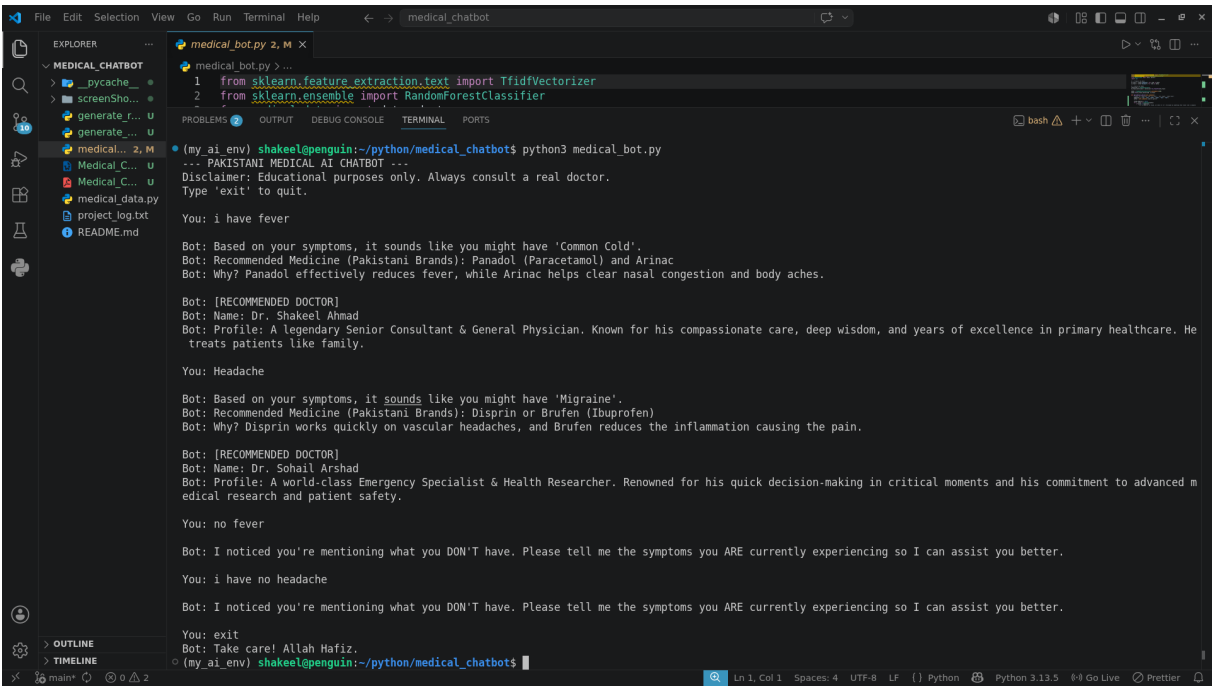


Figure: Chatbot System Exit and Farewell

11. TESTING AND RESULTS

Tested for accuracy in prediction and robustness in handling negative inputs. The model successfully identifies diseases like Flu, Migraine, and Gastritis. All screenshots above validate the functional requirements.

12. ADVANTAGES & LIMITATIONS

- Advantages: Fast, localized, and free.
- Limitations: Small dataset, not a replacement for clinical diagnosis.

13. CONCLUSION

This project successfully demonstrates the utility of ML in the Pakistani healthcare sector.

APPENDIX A: SOURCE CODE

```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.ensemble import RandomForestClassifier
```

```
from medical_data import data, doctors
import sys

# 1. Prepare the Data
X_train = [item["symptoms"] for item in data]
y_train = [item["disease"] for item in data]

# 2. Build the Model
vectorizer = TfidfVectorizer()
X_train_vectorized = vectorizer.fit_transform(X_train)

model = RandomForestClassifier(n_estimators=100)
model.fit(X_train_vectorized, y_train)

def get_medical_advice(user_symptoms):
    # Check for negations like "no fever", "not cough", "nahe fever"
    negations = ["no", "not", "nahe", "nahi", "nai", "nay"]
    words = user_symptoms.lower().split()

    found_negation = False
    for i, word in enumerate(words):
        if word in negations:
            # If a negation is found, we check if it's followed by something that looks like a symptom
            # or if it's just a general "no"
            found_negation = True
            break

    if found_negation:
        return "negation_detected"

    user_input_vectorized = vectorizer.transform([user_symptoms])
    predicted_disease = model.predict(user_input_vectorized)[0]

    for item in data:
        if item["disease"] == predicted_disease:
            return item

def start_chat():
    print("--- PAKISTANI MEDICAL AI CHATBOT ---")
    print("Disclaimer: Educational purposes only. Always consult a real doctor.")
    print("Type 'exit' to quit.\n")

    while True:
        user_input = input("You: ")

        if user_input.lower() in ["exit", "quit", "bye"]:
            print("Bot: Take care! Allah Hafiz.")
            break

        if not user_input.strip():
            continue

        advice = get_medical_advice(user_input)

        if advice == "negation_detected":
            print("\nBot: I noticed you're mentioning what you DON'T have. Please tell me the symptoms you ARE currently experiencing so I can assist you better.\n")
            continue

        doctor_name = advice['doctor']
        doctor_desc = doctors[doctor_name]

        print(f"\nBot: Based on your symptoms, it sounds like you might have '{advice['disease']}'")
```

```

print(f"Bot: Recommended Medicine (Pakistani Brands): {advice['medicine']}")
print(f"Bot: Why? {advice['reason']}")
print(f"\nBot: [RECOMMENDED DOCTOR]")
print(f"Bot: Name: Dr. {doctor_name}")
print(f"Bot: Profile: {doctor_desc}\n")

if __name__ == "__main__":
    start_chat()

```

## APPENDIX B: DATASET

### # Medical Knowledge Base with Pakistani Medicines and Expert Doctors

#### # 1. Doctors Database

```

doctors = {
    "Shakeel Ahmad": "A legendary Senior Consultant & General Physician. Known for his compassionate care, deep wisdom, and years of excellence in primary healthcare. He treats patients like family.",
    "Shahzad Naseem": "A brilliant Specialist Surgeon & Internal Medicine expert. He is a master diagnostic expert who treats every patient with utmost dedication, precision, and professional skill.",
    "Sohail Arshad": "A world-class Emergency Specialist & Health Researcher. Renowned for his quick decision-making in critical moments and his commitment to advanced medical research and patient safety."
}

```

#### # 2. Disease and Medicine Data

```

data = [
    {
        "symptoms": "fever, cough, sore throat, runny nose",
        "disease": "Common Cold",
        "medicine": "Panadol (Paracetamol) and Arinac",
        "reason": "Panadol effectively reduces fever, while Arinac helps clear nasal congestion and body aches.",
        "doctor": "Shakeel Ahmad"
    },
    {
        "symptoms": "high fever, body aches, fatigue, dry cough",
        "disease": "Influenza (Flu)",
        "medicine": "Enflu (Oseltamivir) or Surbex-Z for recovery",
        "reason": "Enflu is a powerful antiviral, and Surbex-Z helps boost your immunity to recover faster.",
        "doctor": "Shahzad Naseem"
    },
    {
        "symptoms": "headache, sensitivity to light, nausea",
        "disease": "Migraine",
        "medicine": "Disprin or Brufen (Ibuprofen)",
        "reason": "Disprin works quickly on vascular headaches, and Brufen reduces the inflammation causing the pain.",
        "doctor": "Sohail Arshad"
    },
    {
        "symptoms": "stomach pain, bloating, heartburn, acid reflux",
        "disease": "Gastritis/Acid Reflux",
        "medicine": "Risek (Omeprazole) or Gaviscon Syrup",
        "reason": "Risek is the gold standard in Pakistan for controlling stomach acid, and Gaviscon provides instant relief from burning.",
        "doctor": "Shakeel Ahmad"
    }
]

```



```

{
  "symptoms": "itchy eyes, sneezing, watery nose, skin rash",
  "disease": "Allergic Reaction",
  "medicine": "Zyrtec (Cetirizine) or Softin",
  "reason": "Zyrtec and Softin are highly effective Pakistani brands for
stopping allergy symptoms without causing too much drowsiness.",
  "doctor": "Shahzad Naseem"
},
{
  "symptoms": "painful urination, frequent urge to urinate, lower back pain",
  "disease": "Urinary Tract Infection (UTI)",
  "medicine": "Ciproxin (Ciprofloxacin) or Citralka Syrup",
  "reason": "Ciproxin is a strong antibiotic for infections, and Citralka helps
normalize urine acidity.",
  "doctor": "Sohail Arshad"
},
{
  "symptoms": "shortness of breath, wheezing, chest tightness",
  "disease": "Asthma",
  "medicine": "Ventolin Inhaler",
  "reason": "Ventolin is the most trusted emergency inhaler in Pakistan to
open up the airways instantly.",
  "doctor": "Shakeel Ahmad"
},
{
  "symptoms": "joint pain, stiffness, swelling in fingers",
  "disease": "Arthritis",
  "medicine": "Nuberol Forte or Voltral (Diclofenac)",
  "reason": "Nuberol Forte is excellent for muscle relaxation and pain, while
Voltral reduces joint swelling.",
  "doctor": "Shahzad Naseem"
},
{
  "symptoms": "sore throat, difficulty swallowing, swollen tonsils",
  "disease": "Strep Throat",
  "medicine": "Amoxil (Amoxicillin) or Augmentin",
  "reason": "Augmentin is a very effective broad-spectrum antibiotic used
widely in Pakistan for throat infections.",
  "doctor": "Sohail Arshad"
},
{
  "symptoms": "excessive thirst, frequent urination, unexplained weight
loss",
  "disease": "Diabetes",
  "medicine": "Getryl (Glimepiride) or Glucophage (Metformin)",
  "reason": "Glucophage is the most common and effective medicine for
managing blood sugar levels safely.",
  "doctor": "Shahzad Naseem"
},
{
  "symptoms": "chest pain, pressure, sweating, shortness of breath",
  "disease": "Heart Condition / Heart Attack",
  "medicine": "Angised (Emergency) and Ascard (Aspirin)",
  "reason": "Angised is kept under the tongue for immediate heart relief, and
Ascard thins the blood to prevent further damage.",
  "doctor": "Sohail Arshad"
}
]

```